

MISP
INTELLIGENCE SHARING

MISP data model overview

Structure, context, relationships and feedback

Team CIRCL
TLP:CLEAR

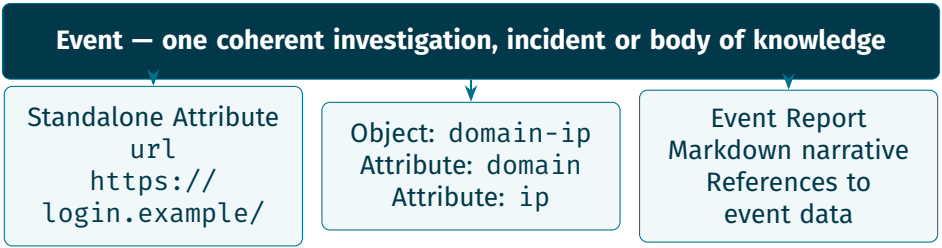
MISP Project
https://www.misp-project.org/

Reference: 7 November 2026 — MISP 2.5

Read the model in four parts

Structure	Event, Attribute, Object, Event Report
Context	Tags, taxonomies, galaxies and clusters
Analysis / feedback	Analyst data, sightings and proposals
Sharing	Organisations, distribution and sharing groups

An event is a contextual envelope



Containment: object attributes also belong to this event. All three branches are optional.

- Event metadata records identity, creator, date and analysis state.
- Distribution and sharing groups help define the audience.

Attributes are typed data points

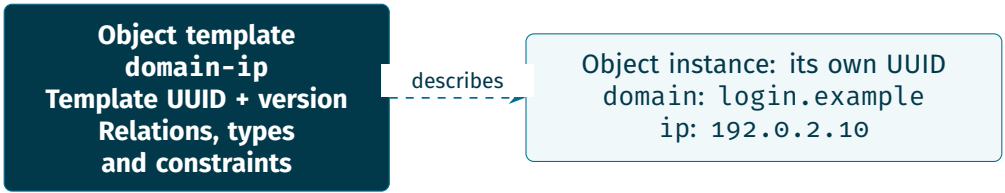
A standalone attribute

```
type:  domain
value:  login.example
category:  Network activity
to_ids:  false
```

- Type gives the value its meaning.
- UUID identifies this attribute.
- Comments and observation times add detail.
- Tags and distribution add context and sharing controls.

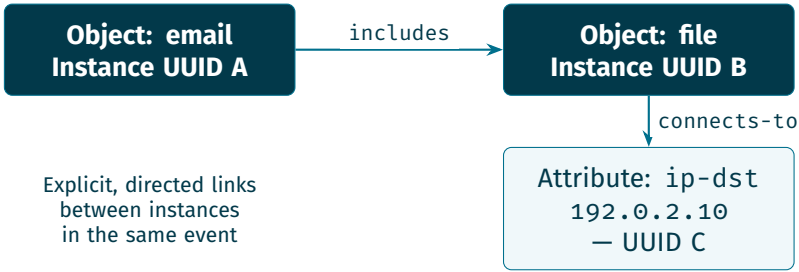
`to_ids` signals intended detection use; assess confidence and freshness separately.

Objects group attributes; templates describe the shape



Model one meaningful observation. Reuse a maintained template before inventing one.

Object references state a directed relationship



- Source: an object instance. Target: an object or attribute.
- Store the relationship type and referenced instance UUID.

Event Reports connect narrative to structured data

- Keep the investigation story in Markdown inside the event.
- Reference attributes, objects and tags using MISP report syntax.
- Review the report's distribution and any sensitive prose.

Illustrative report fragment

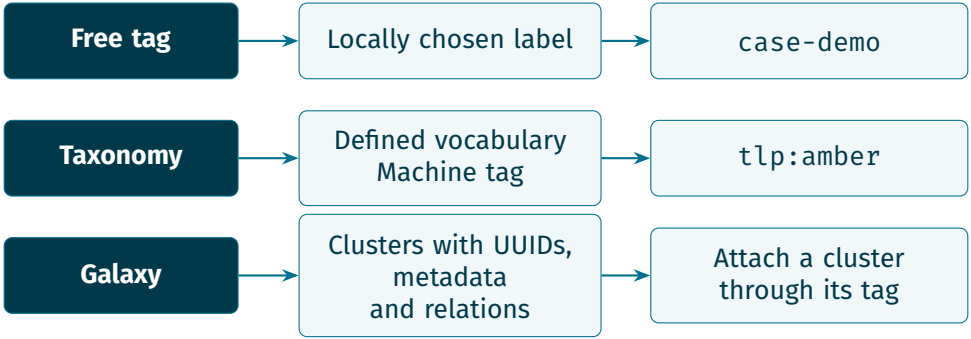
The message led to `@[attribute](<attribute-uuid>)`.
The observation is recorded in `@[object](<object-uuid>)`.

Identity survives sharing through UUIDs

Identifier	Purpose
Local numeric ID	Locates a record on one MISP instance
Element UUID	Identifies an event, attribute, object or other element across instances
Template UUID + version	Identifies the definition used to describe an object
Attribute value	Can match values in other records; does not identify the record

Preserve UUIDs when updating shared records; use new UUIDs for new observations.

Three ways to add context



A taxonomy defines labels. A galaxy organises richer concepts represented by clusters.

Tag scope and handling must be explicit

- Event tags describe the event; attribute tags describe specific data.
- Use the narrowest appropriate scope for confidence and classification.
- Local tag attachments stay local; shareable attachments can travel with data.
- A TLP label communicates handling expectations. Configure access separately.

Avoid an inheritance assumption

Event context can be displayed or used by consumers without being copied into each attribute's tag list.

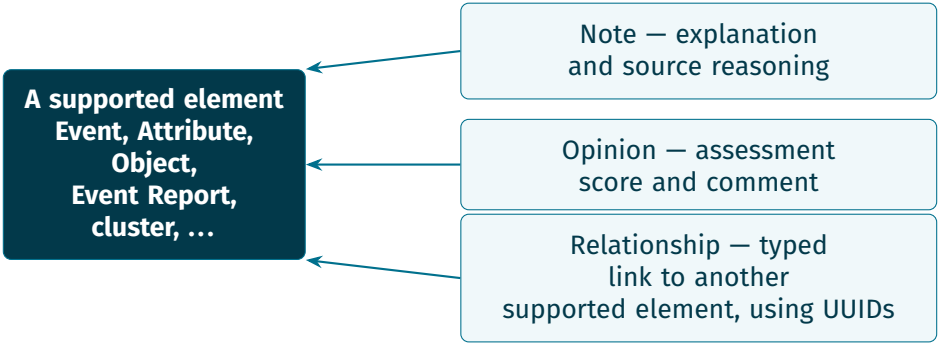
Galaxy clusters enrich classification

- A galaxy groups a domain of knowledge: techniques, actors, tools, …
- A cluster represents one concept with a UUID, description and metadata.
- Cluster relations can connect concepts, including across galaxies.
- Attach relevant clusters to the event or specific attributes.

Example

Link the observed behaviour to a MITRE ATT&CK technique cluster.
Explain the evidence separately; a technique label alone does not establish attribution.

Analyst data preserves reasoning and assessments



Analyst data has its own authorship, UUID and distribution. Arrows indicate attachment.

Sightings and proposals close the feedback loop

Sighting: feedback on an attribute

`0` — sighting / observation
`1` — false-positive report
`2` — expiration

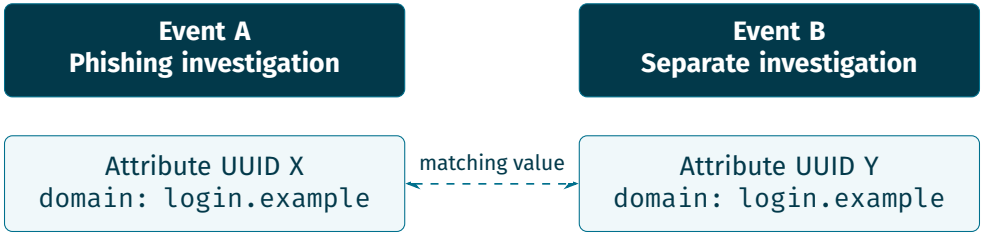
Include time and source. Visibility depends on the configured sighting policy.

Proposal: suggested change

Suggest an attribute addition, modification or deletion for the event owner to review.

Record detailed reasoning in analyst data when appropriate.

Correlation reveals matches to investigate



Correlation depends on types and configuration

Mode	Example	Conditions
Value match	Same eligible domain or hash	Type/engine rules and correlation enabled
CIDR	`192.0.2.0/24` contains `192.0.2.10`	Advanced correlations enabled; supported IP types
ssdeep	Similar fuzzy hashes	Advanced correlations, ssdeep runtime support and configured threshold

Non-correlating types, exclusions and over-correlation limits can suppress matches.

Choose the right kind of link

Mechanism	Connects	Meaning
Object relation	Object to its attribute field	`ip` names a field
Object reference	Object to object / attribute	Explicit directed assertion
Analyst Relationship	Supported elements by UUID	Analyst's typed assertion
Cluster relation	Galaxy clusters	Knowledge-base relationship
Correlation	Eligible matching attributes	Computed investigative pivot

Sharing settings constrain what leaves the event

- Event distribution sets the containing audience; a sharing group can name it.
- Child settings can inherit where supported, or further restrict sharing.
- Check object, attribute, report and analyst-data settings separately.
- Check the actual recipient view and export, including free-text reports.

Keep handling and access aligned

Distribution, sharing groups, permissions and sync/export settings enforce different parts of the workflow. Tags communicate context and handling.

Keep event boundaries meaningful

- Group observations that share an investigation, source or time context.
- Split unrelated incidents rather than building an unbounded indicator list.
- Use an extended event to add your perspective to a parent event.
- The extension retains its own identity, ownership and sharing settings.

Extended event

The child event's `extends_uuid` identifies the parent event.
A combined view depends on the reader being allowed to see the relevant events.

Model a phishing observation

1. Create an event for one investigation and define its audience.
2. Add a `domain-ip` object for the domain and observed address.
3. Add a standalone URL attribute if it has its own investigative role.
4. Describe evidence in an Event Report; attach relevant context tags.
5. Record an analyst Note and later sightings on the relevant attributes.

Use `login.example` and `192.0.2.10` for discussion.
Inspect correlations as leads; review `to_ids` before consumer use.

Exercise: choose an element and explain why

1. A domain was observed resolving to two IP addresses in one observation.
2. Another investigation contains the same domain.
3. A file in this event contacted one of those addresses.
4. A reviewer wants to explain doubts about the attribution.
5. A recipient observed the domain yesterday.
6. The event should be available only to a named community.

Which choices express evidence, which express analysis, and which control access?

A useful record preserves meaning

- Structure observations with attributes and coherent objects.
- Preserve narrative, provenance, UUIDs and observation times.
- Distinguish classification, asserted relationships and computed matches.
- Keep feedback and assessments attached to the right element.
- Verify the audience and payload before relying on sharing.

References and further training

- MISP standard: exchange formats and schemas
- MISP Book: usage, sharing, analyst data and sightings
- MISP Objects: maintained object definitions
- MISP Taxonomies and MISP Galaxy: classification and context
- MISP 2.5.51 source reviewed: model and correlation behaviour

Review date: 7 November 2026. Build instructions, review findings and source links are in the course README.